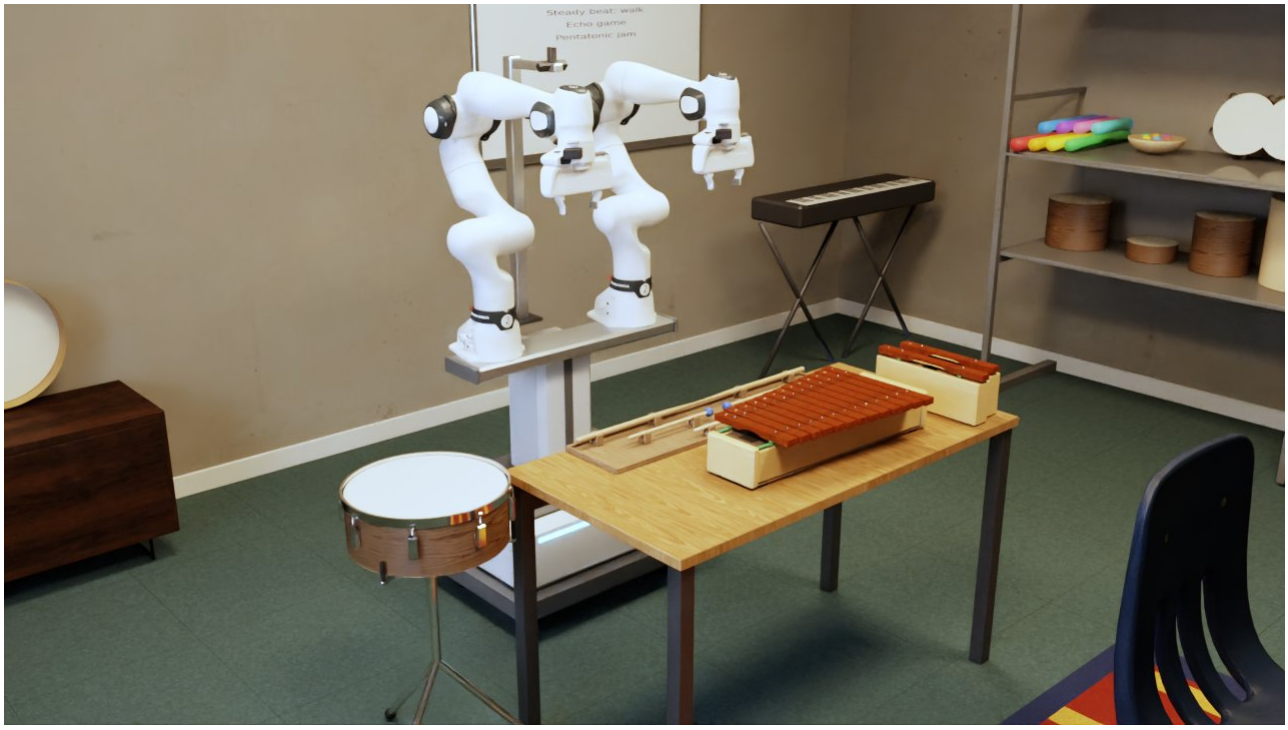
Tap a percussion instrument

The percussion simulation: physics, tasks, validation and runtime

A Newton 1.5 / MuJoCo environment of a music-therapy session: two Franka FR3 arms on a parked mobile platform (or two Pandas, one FR3, or a Unitree G1 humanoid with Dex3 hands) play an Orff alto xylophone, single resonator bars and a 14-inch drum with drumsticks, mallets, a felt beater or their fingertips. Every stroke is solved against the instrument's vibration modes at 48 kHz; the sound radiates into the room and is heard at the client's chair (scoring) and at the robot's microphone (observation). This report contains no autonomous task solution.



The therapy preset, settled at t = 0 (Isaac RTX render of the simulated state; 1280 x 720).

What	Result	Evidence
Bar partials (13 bars)	1st/2nd exact by construction, 3rd within 16.0 %; F4 mass 134.1 g vs 134.9 g measured (not fitted)	bench E1
Drumhead modes	within 2.0 % of Wagner at three tensions (air-loaded)	bench E2
Stick contact on the head	4.00 ms at mid tension (Wagner 3-6 ms); the contact patch is the loaded area's physical size, halved/doubled: contact time within 9.5 %	bench E4, convergence
Numerical refinement 48 vs 192 kHz	every gated observable within 2 % (or one sample)	convergence
Contact friction, room boundaries	a mallet pushed on the head stops in 15 mm (frictionless: glides 71 mm into the hoop); a rolling stick stops within 0.13 m; a thrown stick bounces off the wall	friction_floor
Assisted task episodes (evaluator)	7/7 pass (dynamics, single-tone, echo, steady-beat, accent-ostinato, bordun, roll); strike-and-stop: a phantom-held felt pressed at the cue passes (pass), no stop fails	phantom_tasks
Through the command path (seed 0)	strike-and-stop's stop by the FR3 (pass, 23.23 dB below the free ring); finger-tap by the G1 (pass); take-sticks by the FR3 (pass). The same scripts on other seeds: strike-and-stop seed 1 fail, seed 2 pass, seed 3 fail; finger-tap seed 1 fail, seed 2 fail	strike_and_stop, g1_finger_tap, command_path_seeds
Static friction	constant targets for 3 s with the hand pressed: the G1's fingertips move 0.043 mm on the head, the FR3's pads 0.002 mm on the bar; the client's ear in the last second <= -38.9 / -79.2 dB SPL	pressed_hold
G1 stick strokes p..f	2.0 m/s / 4.06 ms, 2.3 m/s / 4.29 ms, 3.2 m/s / 4.00 ms, 3.8 m/s / 3.85 ms	g1_stick_strokes
G1 16-beat run	16 strikes, one contact each, IOI CV 0.17 %	g1_stick_strokes

What	Result	Evidence
Phantom melody / pentatonic set-up	evaluator pass / pass	phantom_*
FR3 take-sticks	evaluator pass	take_sticks
Rest, 20 s	FR3 presets: 0.0 mm drift; G1: pelvis 1.347 mm, tilt 0.2435 deg	rest, g1_rest
Runtime (public policy path)	FR3 physics 0.9-37.4x; G1 1.5-90.5x; with cameras <= 124x (limit 200x)	runtime

01 / Acceptance gates

The acceptance gates read the same results as this report: 62 of 62 gates pass. Each gate states its criterion, and a failed gate stays listed.

Gate	Criterion	Value	
drumhead modes vs Wagner	max |error| <= 3 %	1.97	pass
F4 bar mass (not fitted) vs Rucz 134.9 g	within 2 %	134.1	pass
mallet contact on F4	0.45-0.8 ms (Rucz 0.6-0.7)	0.542	pass
stick first contact on the head (mid tension, mf)	3-6 ms (Wagner 2006: 3-6 ms)	4.0	pass
elastic energy closure (bar)	|balance| < 1e-3	7.167877402736167e-13	pass
refinement: mallet_on_F4_bar	gated observables <= 2 % (contact time: or one sample)	2.97	pass
refinement: stick_on_drum_mf	gated observables <= 2 % (contact time: or one sample)	0.655	pass
refinement: stick_on_drum_f	gated observables <= 2 % (contact time: or one sample)	0.298	pass
refinement: fingertip_on_drum	gated observables <= 2 % (contact time: or one sample)	0.288	pass
refinement: fingertip_tap_reference	gated observables <= 2 % (contact time: or one sample)	0.28	pass
refinement: drum_mode_truncation	gated observables <= 2 % (contact time: or one sample)	0.157	pass
contact patch halved / doubled (physical size)	contact time within 10 % (E19)	9.46	pass
impact-window guard invariance (2/20/100 mm)	levels within 0.1 dB, peak forces within 2 %	(0.014, 0.16)	pass
FR3 presets at rest 20 s	drift <= 1 mm, silent	0.0	pass
G1 standing hands-off 20 s	pelvis <= 5 mm, tilt <= 2 deg, objects still	(1.347, 0.2435)	pass
layout sweep FR3	every preset x task x seed 0-2 settles	36/36	pass
layout sweep G1	every preset x task x seed 0-2 settles	36/36	pass
FR3 stress	no fault, no noslip fallback	(None, 0)	pass
FR3 stress: every object stays in the room	all final positions inside the walls	(True, 6.583)	pass
G1 stress	no fault, finite state	(None, True)	pass
G1 stress: every object stays in the room	all final positions inside the walls	(True, 5.506)	pass
friction on the head: a pushed mallet stops	stopped within 50 mm (friction off: slides farther)	(15.4, 71.49)	pass
carpet: a rolling stick comes to rest	at rest within 0.5 m	(0.1256, 0.0)	pass
walls: a thrown stick stays in the room	never behind the back wall	-1.5934	pass
a tool lying still on the head is silent (no static pressure)	|mean pressure| < 1e-3 Pa at both microphones; policy L_Aeq(100 ms) < 30 dB(A)	(1.522061838192201e-07, 0.0)	pass
hand surfaces are strikers (G1 fingers/palm on the head)	non-fingertip proxies in contact and the slap logged	(2, 5)	pass
a hand pressed on the head makes no static pressure	|mean pressure| < 1e-3 Pa at both microphones	2.3403763768200404e-05	pass
lifting the pressed hand is no stroke	L_AF max of the 0.4 s lift below the slap's	(41.0, 73.7)	pass
static friction: G1 hand pressed on the drumhead, constant targets 3 s	contact points move < 0.05 mm in the last second; client's ear < 20 dB SPL in its last second	(0.0001, -38.9)	pass
static friction: FR3 pads pressed on the C4 resonator bar, constant targets 3 s	contact points move < 0.05 mm in the last second; client's ear < 20 dB SPL in its last second	(0.0006, -79.2)	pass
lifting the pressed G1 hand slowly is quiet	2 s lift: L_AF max < 60 dB(A)	{'0.4s': 46.49409866333008, '2s': 51.52964782714844, '0.4s_targets_every_5ms': 75.08653259	pass

Gate	Criterion	Value	
resonator-bar opening sensitivity reported	shipped case and the spread over authored openings	{'C4': {'shipped': 0.617, 'span': (0.617, 0.677), 'slot_share': 0.143}, 'G4': {'shipped':	pass
command-path scripts beyond their seed (reported, not gated on success)	every seed run and recorded	{'strike_and_stop': {'1': False, '2': True, '3': False}, 'g1_finger_tap': {'1': False, '2'	pass
assisted episode: dynamics	evaluator success	True	pass
assisted episode: single-tone	evaluator success	True	pass
assisted episode: echo	evaluator success	True	pass
assisted episode: steady-beat	evaluator success	True	pass
assisted episode: accent-ostinato	evaluator success	True	pass
assisted episode: bordun	evaluator success	True	pass
assisted episode: roll	evaluator success	True	pass
strike-and-stop attainable by a physical touch (felt pressed at the cue passes, no stop fails)	felt press: success; none: stopped false	(True, False)	pass
a strike after the cue that stays on the bar fails (contact open at evaluation)	still open at evaluation and no_strike_after_cue false	(True, False, -1.882)	pass
strike-and-stop by the FR3 through the command path, assessment seed 0 (phantom strike, fingertip mute at the cue; other seeds: command_path_seeds)	evaluator success	(True, 23.23)	pass
strike-and-stop latency budget (two pads at 5 N at the centre)	passes at every exchange phase for touches -50..+50 ms around the cue; no touch: 0 dB	(-0.1, 0.15, 0.0)	pass
a rim stroke is heard (rigid-part click)	>= 1 click on the hoop	2	pass
assisted: phantom melody	evaluator success	True	pass
assisted: phantom pentatonic set-up	evaluator success	True	pass
assisted: FR3 take-sticks	evaluator success	True	pass
G1 stick: clean beat	>= 15/16 beats one contact, IOI CV <= 3 %	(16, 0.166)	pass
G1 stick: held by contact after the fixture	held at the end	right	pass
G1 stick: dynamic ladder	>= 3 dynamics with contact	4	pass
G1 finger taps (finger-tap through the command path, therapy seed 0; other seeds: command_path_seeds)	evaluator success	(True, [72.63, 70.66, 71.25, 70.36, 69.65, 69.63, 70.0, 69.92], 70.47)	pass
assisted bordun: recorded and rendered	evaluator success and a video	(True, 'assisted_bordun.mp4')	pass
assisted roll: recorded and rendered	evaluator success and a video	(True, 'assisted_roll.mp4')	pass
robot configurations	dual-fr3, single-fr3, dual-panda, g1 build and move	{'dual-fr3': True, 'single-fr3': True, 'dual-panda': True, 'g1': True}	pass
runtime on the public path	every measured regime < 200x wall/sim (58 rows, 7 files)	123.88	pass
G1 both hands pressed on the head, with and without the head camera	< 200x, at the live floor too	[90.51, 66.18, 84.98, 106.12]	pass
runtime over HTTP (gar run, policy key, live floor)	every row < 200x	[5.33, 30.7, 79.55, 9.22, 46.26, 60.22]	pass
expensive regimes over HTTP, policy key, live floor: series average	every series < 200x wall/sim incl. the capability and image GETs	{'g1_nocam_pressed': 79.6, 'g1_headcam_pressed': 99.7, 'fr3_3cams_resting': 103.7}	pass
expensive regimes over HTTP: floor-length commands over 200x (the floor is a forecast)	at most 10 % of each series' POSTs over 200x (worst command reported)	{'g1_nocam_pressed': ('0/30', 103.4), 'g1_headcam_pressed': ('1/30', 229.6), 'fr3_3cams_re	pass
gar sessions under both keys (FR3, G1)	every check passes	all	pass
evaluator adversarial tests	all pass	49/49	pass

02 / The activity and the tasks

Four task rows from two O*NET occupations (Preschool Teachers: music activities with a drumstick; Music Therapists: sounds for assessment, tempo and harmony, playing percussion to keep a rhythm).

A music therapist's session uses a small set of instruments chosen for clear, controllable sound: an Orff alto xylophone (the instrument whose bars Rucz et al. measured bar by bar), single resonator bars, and a hand or snare drum played with sticks or a beater. The environment's four presets are small setup changes of one room: a xylophone session (therapy), a gait-training beat with the drum alone (gait), an assessment table with two resonator bars, the drum and a loudspeaker (assessment), and a pentatonic set-up where the F and B bars come off (pentatonic). Tasks are judged on contact events and the simulated sound at the client's ear; level targets come from reference strokes replayed through the same model, not hand-set numbers.

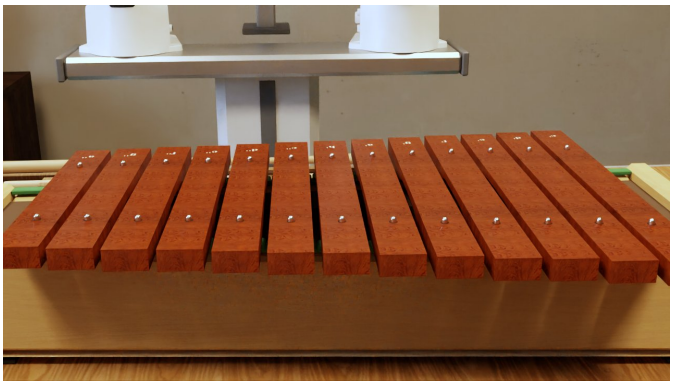
Preset	Tasks (seeded goals)	What the robot must do
therapy	melody, bordun, roll, finger-tap	play a tune / an alternating bordun with mallets, a roll, fingertip taps
gait	take-sticks, steady-beat, accent-ostinato	take both sticks, keep a walking cadence, accented ostinato
assessment	single-tone, dynamics, echo, strike-and-stop	one clean tone, p-mf-f ladder, echo a heard rhythm, stop the ring on cue
pentatonic	pentatonic-setup	lift the F and B bars off their pins into the tray



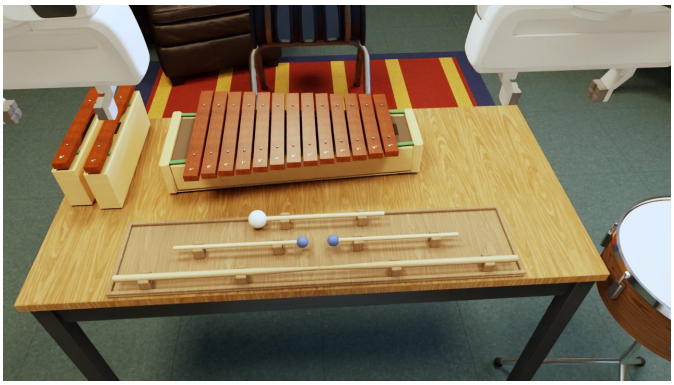
Gait preset: the drum alone between two trap tables.



Assessment preset: two resonator bars, the drum and the loudspeaker that plays the echo stimulus.



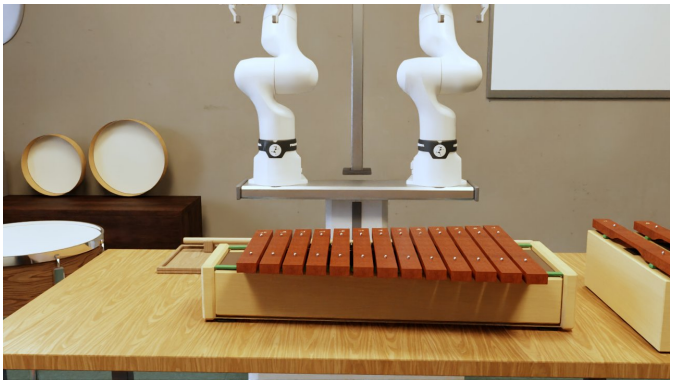
Pentatonic preset: the bar tray beside the xylophone.



Policy camera (station mast, D435-class, 69 deg HFOV).

03 / Scene and robots

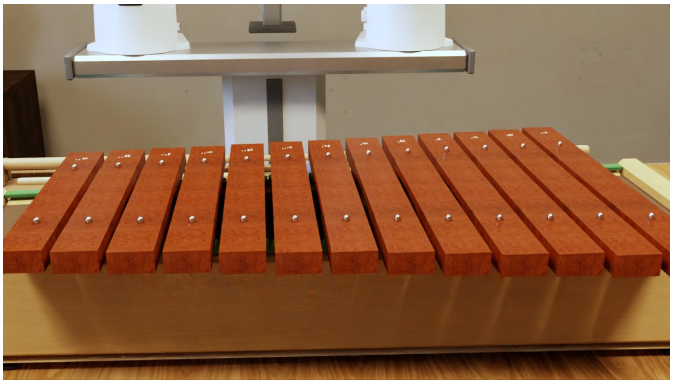
Dimensions are the named products' (Studio 49-class Orff alto xylophone, Remo coated 14 x 5 in drum, Vic Firth 5A sticks); materials are rendered from pinned CC0 textures. Visual-only dressing is separate from colliders (dressing.py vs scene.py).



From the client's chair: xylophone (rosewood bars on felt and pins over the box resonators), resonator bars, the drum; the FR3 pair on the parked platform.



Side view: the beech stick tray with sticks, mallets and the felt beater in saddles; the drum on its stand.



Xylophone close-up: graduated bars, felt strips on the rails, end blocks.



Drum close-up: walnut shell, chrome triple-flanged hoop, eight lugs, coated head.



The instrument shelving, stocked (render-only): hand drums, a djembe, boomwhackers, egg shakers, tambourines, frame drums, song binders, bongos, chime bars. Walls and every piece of furniture also have box colliders (params.FURNITURE).

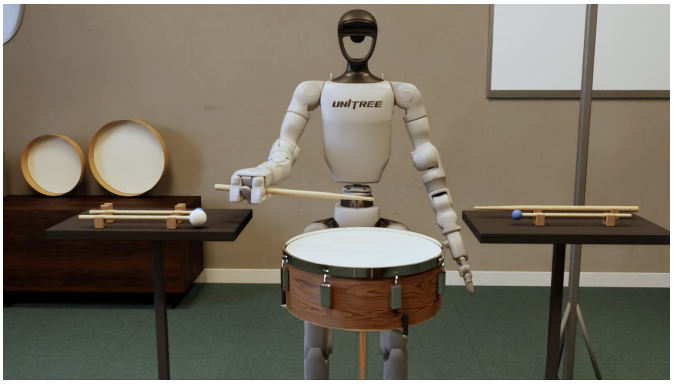


The snare stand's basket arms reach past the shell and hold it with rubber-sleeved hooks.

Configuration	Robot	Control	Qualified evidence
dual-fr3 (default)	2 x FR3 + Franka Hand on the platform beam, bases 0.80 m	joint/Cartesian targets through libfranka rate limits (10 rad/s^2, 5000 rad/s^3); force-mode grasp; collision reflexes	all FR3 experiments
single-fr3	right FR3 only	same	robots smoke
dual-panda	2 x Panda	Panda per-joint limits	robots smoke
g1	Unitree G1 29-DoF + two Dex3-1 hands, floating, standing	PD joint targets (URDF speeds), waist/arm gravity feed-forward, 0.5 ms steps; Dex3 fingertip strikers; no reflexes	g1_rest, g1_finger_tap, g1_stick_strokes, g1_slowmo, layout_g1

04 / The humanoid configuration (G1)

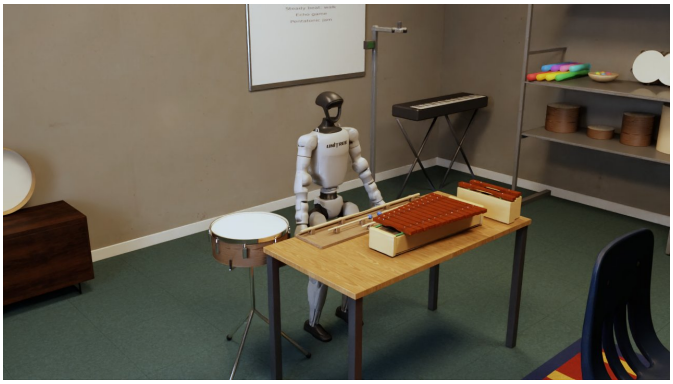
Rig percussion-g1, module g1.py. Floating base, no weld, no balance controller; it stands on its PD servos (the barcode_scanning G1 servo profile, reused).



G1 behind the drum (gait preset) with a stick in its right Dex3 hand (the stick placement into the open hand is an assisted fixture; the hand holds it by contact alone).



Close view of the same state.



G1 at the instrument table (therapy): the drum stands at its right hand.



The G1's head camera (policy observation).

The G1 stands at its preset placement (therapy, assessment and pentatonic: 0.25 m from the table edge with the drum moved to its right hand, 2 cm clear of the table; gait: behind the drum). Its microphone is the head array (one capsule at the head), its policy camera the torso D435. MuJoCo steps the rigid world at 0.5 ms (two substeps per 1 ms tick: the G1 falls at 1 ms, as in barcode_scanning); the impact windows are unchanged (48 x 20.8 us). Each Dex3 index, middle and thumb tip carries a sphere proxy registered as a vibro-contact striker with the rubber-pad law, and the palm and every Dex3 link carry capsule proxies, so any hand surface on an instrument is heard and damps it; the links also collide with their URDF convex hulls. The layout sweep built every G1 preset/task for seeds 0-2: 36/36 settled with the robot up.

05 / Physical model

modes.py, fem.py, modal.py (instrument modes and banks), contact.py and world.py (impact windows), acoustics.py (radiation and room). Parameters with their bases: params.py.

Instruments as modal structures. Each bar is a 1-D Timoshenko beam with its measured undercut profile (fem.py) whose bending and torsion modes below 20 kHz are kept; the xylophone's box resonators couple to their bars through a stiffness term K_{extra}. The drumhead is a tensioned film with bending stiffness, loaded by the air in the shell and outside (the measured frequencies need it); 592 modes below 5 kHz are kept and the rest enter as a residual compliance at the contact point. The drumstick's own three bending modes are a third modal bank. Each bank advances exactly per audio sample: $z_{k+1} = e^{lambda\ dt}\ z_k + G\ f_k$, $G = (e^{lambda\ dt} - 1)/lambda$.

Impact as an implicit per-sample solve. Near a contact the rigid world steps at 48 kHz inside the 1 ms tick (impact window). Per sample, the contact force F solves the nonlinear complementarity of the Hunt-Crossley law $F = K\ d^a(1 + lambda\ d\text{-dot})$ with the structures' and striker's response to that same F: $d = d_{free} - S\ F$, $S =$ Sum c G w (structure) + $c_{res} + dt^2/(2\ m_{eff})$ (striker). m_{eff} comes from the rigid body's or robot's inertia along the contact normal ($J\ M^{-1}\ J^T$). Using the force from the previous sample instead loses 15 % of the energy on a bar and diverges on the drum (M3).

Tangential contact, hand surfaces, rim clicks, room. Every vibro-contact carries regularised Coulomb friction on the solved normal force (mu from 0.3 felt and plastic, 0.35-0.40 hickory - PET film datasheet COF 0.40 - to 0.7-0.8 rubber), applied to both bodies with its torques; a sticking contact is held by a tangential spring to an anchor on the surface (Mindlin's no-slip stiffness, 0.7 of the normal one, critically damped), because a viscous 'sticking' law lets a pressed hand creep. Besides the tools and fingertips, every robot surface that can meet an instrument from above is a striker: the Franka hand housing, link 7 and finger boxes ('plastic') and the G1 palms and all Dex3 links (rubber skin), as capsule proxies fitted to their collision geometry. A rigid contact of a tool or robot link with the hoop, stand, pins, shell, xylophone frame or a resonator box radiates a short transient (energy 1e-3 of the impact's, an estimate), so the robot hears a rim hit. The floor has rolling resistance (carpet: b = 1 mm), the walls and furniture are colliders. The contact patch (the radius over which mode shapes are averaged) is the loaded area's physical size: the film wrapping the striker at its reference force, smoothed by the film's bending length (2.0-2.8 mm for the stick tip).

Sound. Each mode's velocity radiates through a monopole/dipole radiation efficiency to the client's ear and the robot's microphone with the direct path, first reflections and a late reverberation fitted to the room's T60 (0.3-0.8 s from the surfaces' absorption); the evaluator reads L_{AF} (IEC 61672 fast, A-weighted) at the client. Only the oscillating part of each mode radiates, so the quasi-static deflection under a resting load makes no constant pressure. Absolute loudness is not calibrated; one absolute check (bench e8): a 10 m/s accent peaks at 130.29 dB at a seated player's head against 130.8 dB L_peak measured there for a snare with the snares off (Jaroszewski, Rogowski & Rakowski 2000).

Quantity	Value	Basis
Bar wood	rosewood rho 1116 kg/m^3, E_L 24 GPa with Rucz's per-bar orthotropic scale; undercut solved so the 1st and 2nd partials are the measured ones	Rucz 2021 Tables I, II, Sec. III A
Bar undercut	per-bar length / flat / minimum thickness	Rucz 2021 Table II
Drum film	Mylar 0.25 mm, tensions low/mid/high from Wagner's frequencies	Wagner 2006
Stick	Vic Firth 5A hickory, 406 mm, 50 g, three bending modes 390/933/1631 Hz	maker data; M4
Contact laws	hickory/Mylar, rubber/rosewood (Rucz's literature constant), felt, rubber pad (lambda from rubber tan delta 0.15), plastic; friction mu per pair	Rucz; Hertz; estimates within literature ranges
Contact patch	film wrap at the reference force + bending length (stick tip 2.0-2.8 mm)	derived; E19 sensitivity
Resonator-bar boxes	per note: C4 0.320 x 0.05 x 0.11 m, wall 3.4 mm, 1390 cm^3, slot 25 x 44.9 mm; G4 0.288 m, 4.0 mm, 1200 cm^3, 25 x 121.6 mm (tuned to f1 with the slot's own end correction; G4 past the lumped model)	Studio 49 KB/AX datasheet; walls from the mass; slot solved
Finger-tap reference	a fingertip tap: 21.5 g at 0.5 m/s, rubber-pad law (one 7.5 ms contact)	Hajian & Howe 1997; Rempel et al. 1994

Quantity	Value	Basis
Room	6 x 5 x 2.75 m, carpet tiles, gypsum walls, mineral ceiling; T60 0.29-0.82 s	Owens Corning table (walls); handbook tables (floor, ceiling, window)
FR3 limits	velocity 2.62-5.26 rad/s, acceleration 10 rad/s^2, jerk 5000 rad/s^3	Franka FR3 datasheet
G1 servos	kp/kd per joint group, URDF efforts; Dex3 1.4-2.45 N m	G1 robot model, barcode_scanning servo profile
Floor, walls	carpet: Coulomb 0.6 + rolling resistance 1 mm; walls and furniture box colliders	wheelchair coasting tests (rolling); pinned models' bounding boxes

06 / Tasks and their evaluation

task.py: every check reads the contact log, rigid poses and the simulated sound at the client's ear; level targets are reference strokes replayed through the same model.

Task	Preset	Success (all of)
take-sticks	gait	both sticks held >= 5 s with the tips 3-15 cm above the head; no strike; both back on their rests
steady-beat	gait	32 beats at the cadence then 16 at +10 %: count, tempo within 1 %, CV <= 3 %, clean, level SD <= 2 dB, no rim
accent-ostinato	gait	8 bars of 4: accents >= 6 dB above the rest, CV <= 4 %, clean, no rim
melody	therapy	notes in order on the fitted beat grid (50 ms), clean, tone, no neighbour hit
bordun	therapy	8 pairs, both bars within 15 ms, levels within 3 dB, steady tempo
roll	therapy	>= 8 strokes/s over 2 s, alternating mallets, approach speeds within 3 dB SD (not the phase-dependent energy added)
finger-tap	therapy	8 fingertip taps at the tempo, clean, within 4 dB of the fingertip-tap reference (21.5 g at 0.5 m/s)
single-tone	assessment	one clean mf stroke on the bar, mode-1 dominant, within 3 dB of the reference
dynamics	assessment	p, mf, f on the drum, each within 3 dB of its reference and >= 6 dB above the last
echo	assessment	echo the loudspeaker's rhythm: same count, start within 2 beats, interval ratios within 10 %
strike-and-stop	assessment	a tone of stated length: one strike on the C4 resonator bar; not damped before the cue; 0.327-0.427 s after the cue (one bar-box exchange period) the bar >= 10 dB below its exact free ring; no energy-adding contact after the cue
pentatonic-setup	pentatonic	F4, B4, F5 in the tray; the other ten seated on their pins

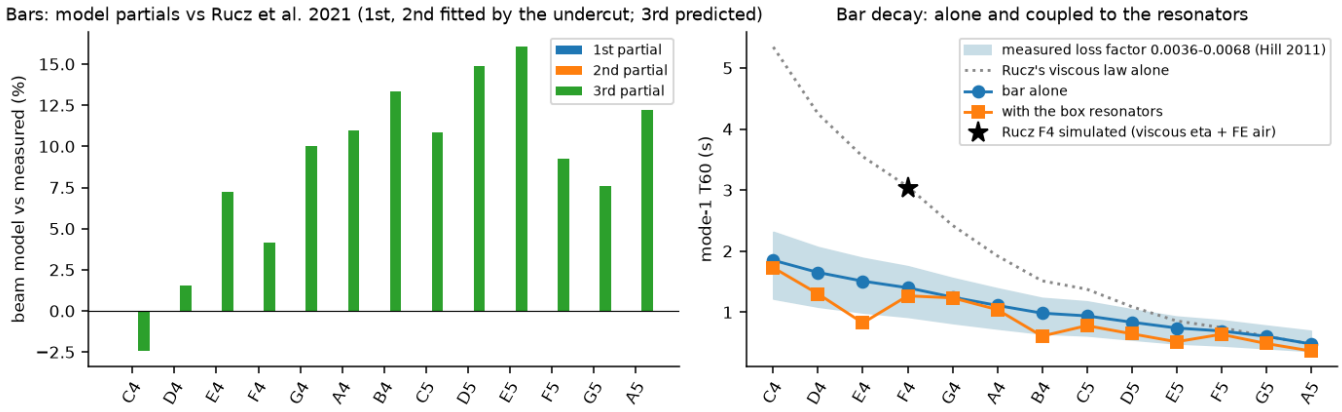
Adversarial evaluator tests: 49/49 episodes behave as expected - synthetic contact logs, level and energy traces, plus three strike-and-stop episodes recorded from the physics and replayed (no stop, a phantom-held felt pressed at the cue, the FR3 mute). The synthetic ones carry the faults each check exists for (double bounce, buried stroke, rim stroke, drifting tempo, wrong note, touched neighbour vs a still-ringing note, node stroke, asynchronous bordun, one-handed roll, too loud, unordered dynamics, reversed echo, undamped bar, weak or late mute, damping before the cue, an anticipated touch just before the cue, strike after the cue, an episode ending before the judgement, stick instead of fingertip, arm still moving) and the exact episodes that must pass.

Case	Expected	Observed	
steady-beat: exact	success	success	pass
steady-beat: one double bounce 40 ms after a stroke	fail: clean	fail: clean, count	pass
steady-beat: first section 3 % slow	fail: tempo	fail: tempo	pass
steady-beat: a rim stroke	fail: no_rim	fail: no_rim	pass
steady-beat: pressed (buried) strokes	fail: clean	fail: clean	pass
steady-beat: alternating 6 dB levels	fail: even_level	fail: even_level	pass
accent-ostinato: accents +8 dB	success	success	pass
accent-ostinato: accents only +3 dB	fail: accents	fail: accents	pass
melody: correct	success	success	pass
melody: one wrong note	fail: notes	fail: notes	pass
melody: mallet also hits the neighbouring bar (20 %)	fail: no_neighbours	fail: no_neighbours	pass
melody: previous note still ringing is not a neighbour hit	success	success	pass
melody: one note 90 ms late	fail: timing	fail: timing	pass
melody: one bar struck over its node (dull)	fail: tone	fail: tone	pass
melody: a neighbouring bar touched and still held at the end (contact open: an extra, buried note)	fail: notes, clean	fail: clean, notes	pass
bordun: 8 pairs 8 ms apart	success	success	pass
bordun: 28 ms asynchrony	fail: together	fail: together	pass
roll: 10 strokes/s alternating	success	success	pass

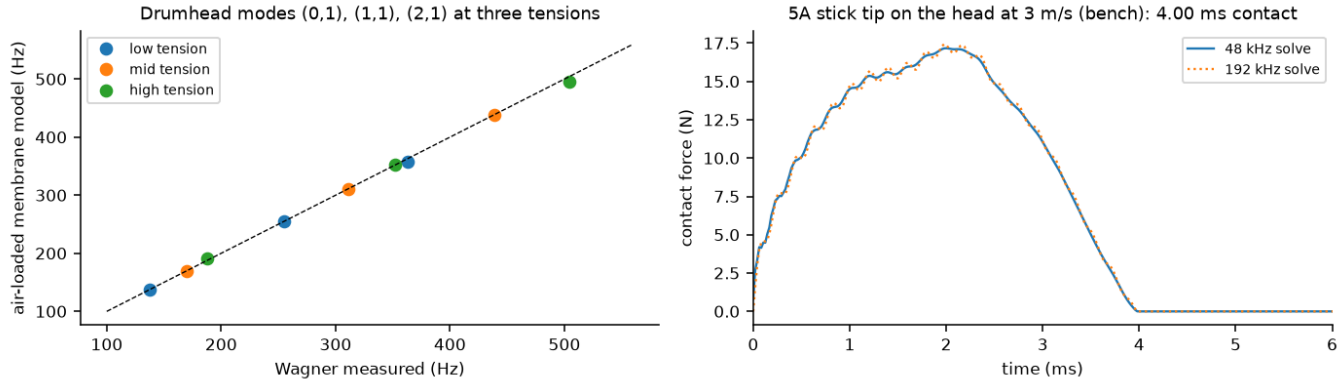
Case	Expected	Observed	
roll: one hand only	fail: alternating	fail: alternating	pass
roll: 5 strokes/s	fail: rate	fail: rate	pass
roll: same mallet speeds, added energy uneven by the bar's phase (passes)	success	success	pass
roll: one hand three times faster than the other	fail: even	fail: even	pass
single-tone: one clean mf tone	success	success	pass
single-tone: second bounce 50 ms later	fail: one_stroke	fail: one_stroke, clean	pass
single-tone: 5 dB too loud	fail: level	fail: level	pass
single-tone: a second bounce that stays on the bar (contact still open)	fail: one_stroke	fail: one_stroke, clean	pass
dynamics: p < mf < f at the references	success	success	pass
dynamics: f before mf	fail: ordered	fail: levels, ordered	pass
echo: same rhythm one beat later	success	success	pass
echo: one note missing	fail: count	fail: count	pass
echo: rhythm reversed (2-1 instead of 1-1)	fail: rhythm	fail: rhythm	pass
strike-and-stop: touch at the cue draining 60 dB/s more than the free ring	success	success	pass
strike-and-stop: not damped	fail: stopped	fail: stopped	pass
strike-and-stop: an anticipated touch landing 20 ms before the cue is no second stroke	success	success	pass
strike-and-stop: damped too weakly (+20 dB/s: 8.6 dB by the window)	fail: stopped	fail: stopped	pass
strike-and-stop: touch 0.3 s late	fail: stopped	fail: stopped	pass
strike-and-stop: damped before the cue	fail: rang_until_cue	fail: rang_until_cue	pass
strike-and-stop: struck again after the cue	fail: no_strike_after_cue	fail: no_strike_after_cue	pass
strike-and-stop: episode ended before the judgement window	fail: stopped	fail: stopped	pass
strike-and-stop: a touch that lifts the bar's energy 15 % as it lands is no second stroke	success	success	pass
strike-and-stop: struck after the cue and left on the bar (contact still open)	fail: no_strike_after_cue	fail: no_strike_after_cue	pass
finger-tap: 8 fingertip taps	success	success	pass
finger-tap: with a stick	fail: fingertip	fail: fingertip	pass
finger-tap: the hand housing left on the head at the end (contact open)	fail: fingertip	fail: clean, count, fingertip, tempo, level	pass
any task: arm still moving at the end	fail: robot_still	fail: robot_still	pass
recorded episode: strike-and-stop, phantom strike, felt press at cue	success	success	pass
recorded episode: strike-and-stop, phantom strike, felt strike after cue and stay	fail:	fail: no_strike_after_cue	pass
recorded episode: strike-and-stop, phantom strike, FR3 fingertip mute (command path)	success	success	pass
recorded episode: strike-and-stop, phantom strike, no stop	fail:	fail: stopped	pass

07 / Bench validation against measurements (no robot)

The package's own mode tables, modal banks and contact solve, checked against the published data they are based on.



Bars (E1): model partials against Rucz's measured ones and mode-1 decay alone / with the resonators.

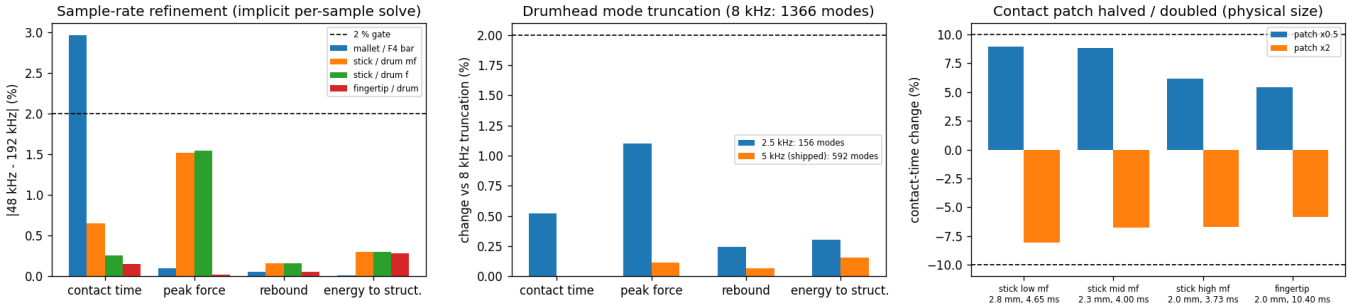


Drum (E2, E4): air-loaded membrane frequencies against Wagner at three tensions; the bench stick stroke's force at the shipped 48 kHz and at 192 kHz.

Check	Model	Literature
Mallet on F4, centre: contact	0.54 ms, mode-1 share 1.0	0.6-0.7 ms (Rucz)
Mallet over the node	mode-1 share 0.0 (fundamental suppressed)	node strike suppresses it
Stick on head, mf	4.00 ms, 17.2 N (patch 2.29 mm)	3-6 ms (Wagner 2006), 4.5-8 ms in total (Dahl 2011); 1.8-107 N
Absolute level (e8)	f stroke 125.84 dB peak, 10 m/s 130.29 dB, at a seated player's ear (0.592 m)	130.8 dB L_peak at the player's head, snare with snares off, practice selections (Jaroszewski, Rogowski & Rakowski 2000, Table 1) - like for like with the 10 m/s accent; 134.8 dB at the ear canal, moderate force (Valderrama et al. 2025) - a lower bound only
Elastic energy closure	48 kHz: 1.8e-13, 96 kHz: 7.2e-13, drum 48 kHz (undamped head + residual compliance): 6.6e-03	|balance| < 1e-3

08 / Numerical refinement

Gate: <= 2 % change at the shipped settings against the finest (contact time: or one 48 kHz sample).



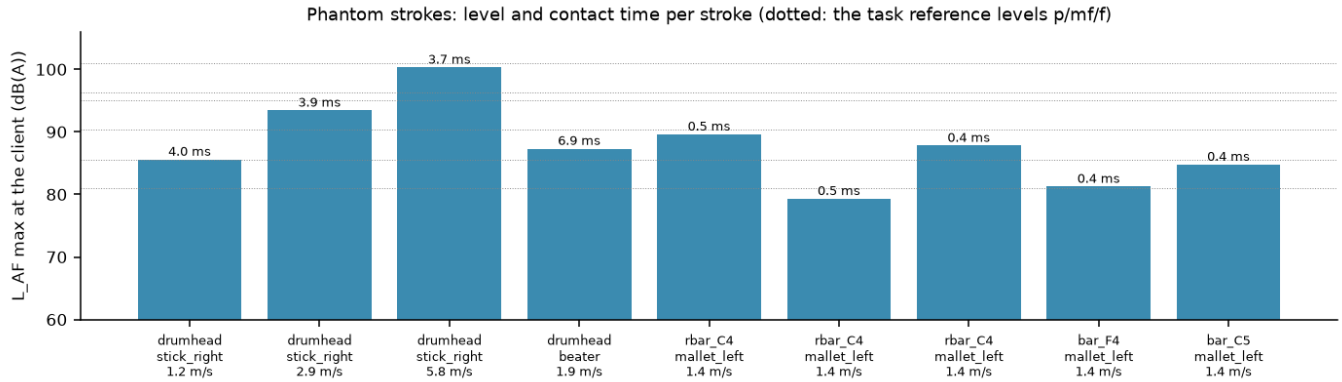
Left: the same strokes at 48 (shipped), 96 and 192 kHz. Middle: the head's modal truncation at 2.5, 5 (shipped) and 8 kHz with the residual compliance in series. Right: the contact patch (the loaded area's physical size) halved and doubled; gate 10 % on the contact time (E19).

Case	contact time	peak force	rebound	energy to structure	gate
mallet on F4 bar	+2.97 %	-0.10 % (not gated)	+0.054 %	-0.013 %	pass
stick on drum mf	+0.66 %	-1.52 % (not gated)	-0.163 %	-0.304 %	pass
stick on drum f	+0.26 %	-1.54 % (not gated)	-0.164 %	-0.298 %	pass
fingertip on drum	+0.15 %	-0.03 % (not gated)	-0.060 %	-0.288 %	pass
fingertip tap reference	+0.28 %	-0.04 % (not gated)	-0.068 %	-0.274 %	pass
head modes 5 vs 8 kHz	+0.00 %	-0.12 %	-0.065 %	-0.157 %	pass

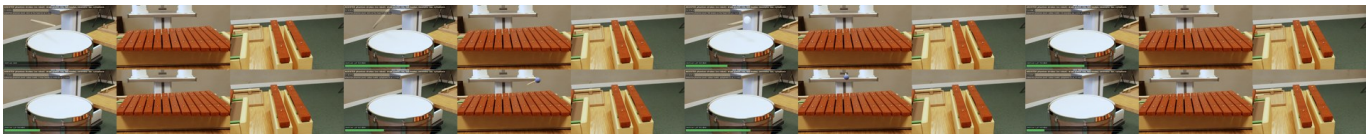
The mallet's 0.54 ms contact is 26 samples, so its 3 % change is under one sample. The sampled peak force of a sub-millisecond pulse converges more slowly (about 1.5 %) and is reported, not gated; the energy that sets the loudness changes by 0.3 % (0.01 dB). The finger-tap reference is a fingertip tap (21.5 g at 0.5 m/s, gated). The window guard (when a contact window opens) changes no outcome: window_invariance. **Contact patch:** halving or doubling it changes the stick's contact time by at most 9.5 % and the energy left in the head by about 10 % (0.5 dB); the shipped stick contacts span 3.71-4.67 ms (Wagner 3-6 ms; Dahl 4.5-8 ms in total, with re-contacts this bench does not show).

09 / Assisted strokes and melody (phantom fixtures)

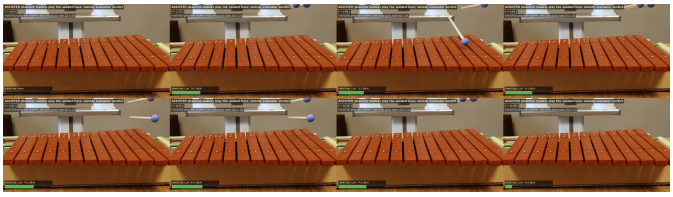
A phantom spring holds a tool at its grip point and launches the stroke; the robots stay at ready. These show the instruments' interactions, not a policy's ability.



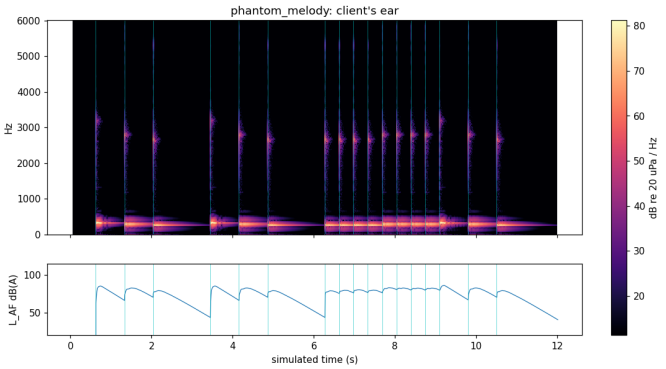
Phantom strokes on the drum and bars: every stroke one contact, levels near the task references; the rim shot excites nothing and the node strike is duller.



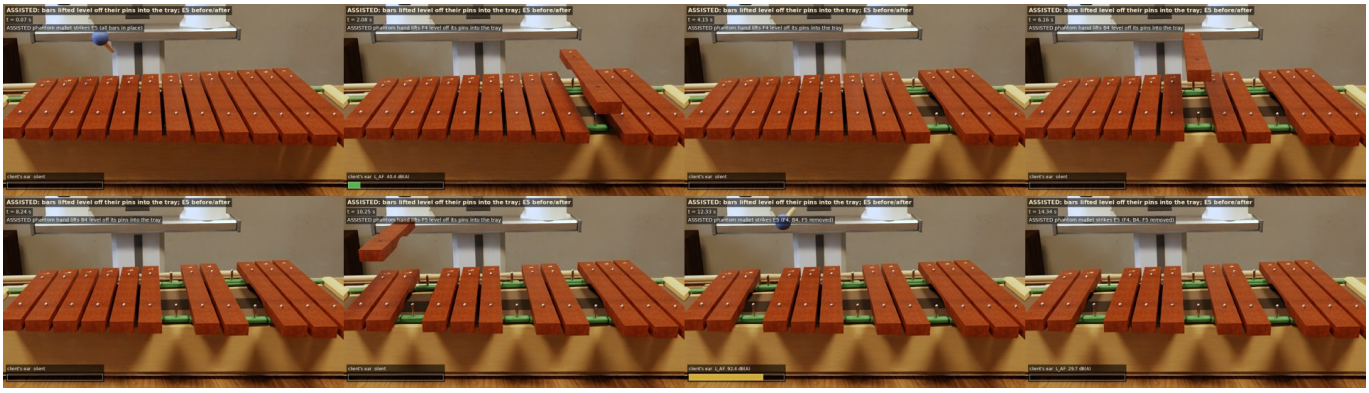
Phantom strokes, contact sheet of the video (with sound).



Phantom two-mallet melody (Hot Cross Buns), evaluator pass.



Its spectrogram at the client: the bars' partials and decays.



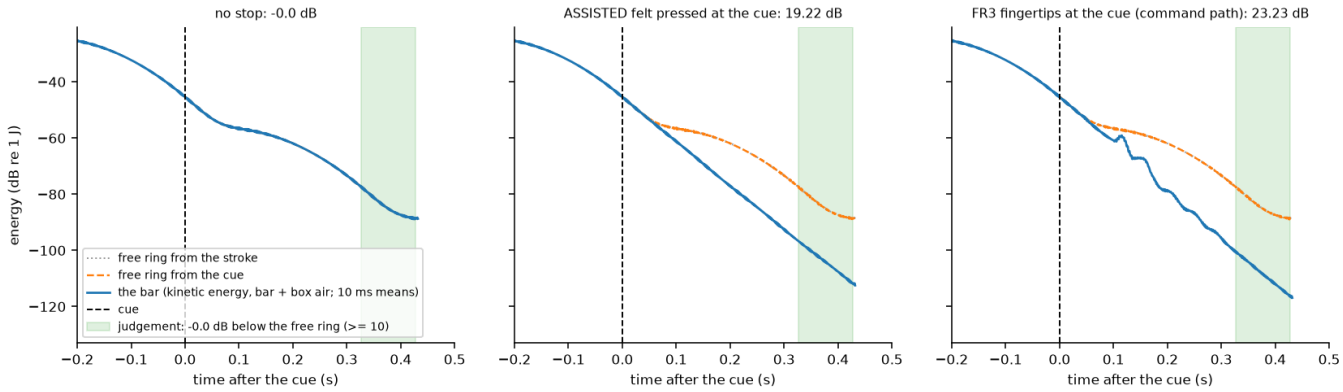
Pentatonic set-up by phantom fixtures (bars lifted off their pins into the tray), evaluator pass.

10 / Assisted task episodes through the evaluator

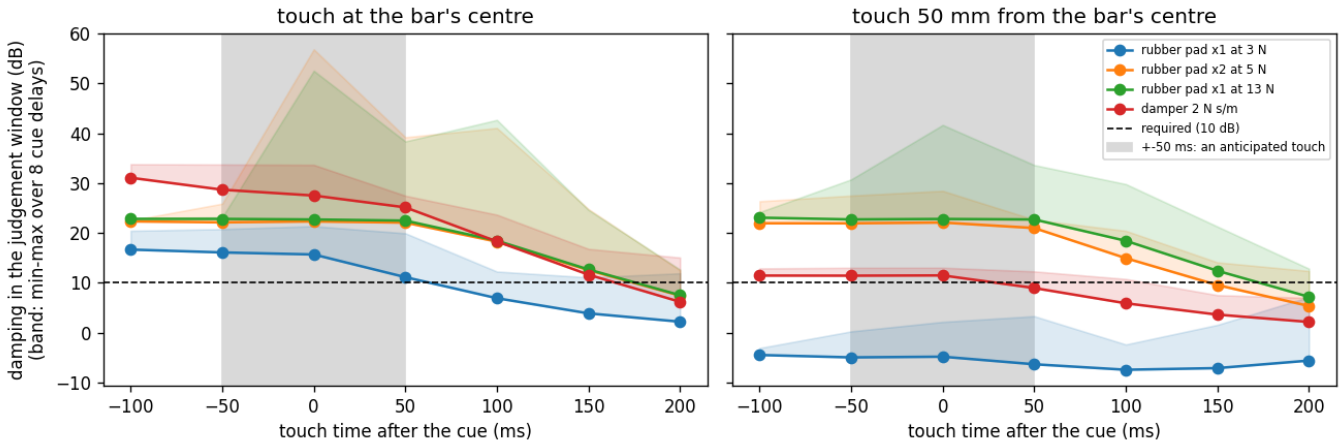
Phantom pivots play each task and the shipped evaluator judges them. Assistance: the tools are held and launched by phantom springs; contacts, sound and scores are the environment's.

Task	Episode	Evaluator	Failed checks
dynamics	stick p, mf, f on the drum	pass	-
single-tone	rubber mallet, mf, on the seeded resonator bar	pass	-
echo	the loudspeaker's rhythm echoed on the drum	pass	-
steady-beat	48 stick strokes, the last 16 at +10 %	pass	-
accent-ostinato	32 strokes at 116 BPM, every 4th forte	pass	-
bordun	8 simultaneous mallet pairs on the xylophone	pass	-
roll	two mallets alternating at 8 strokes/s on the C4 resonator bar	pass	-
strike-and-stop	phantom strike, nothing stops it	fail	stopped
strike-and-stop	phantom strike, ASSISTED: a phantom-held felt set down on the bar at the cue	pass	-
strike-and-stop	phantom strike, the same felt set down ~0.1 s after the cue	fail	stopped
strike-and-stop	the felt STRIKES the bar ~30 ms after the cue and stays on it (contact still open at the evaluation)	fail	no_strike_after_cue
strike-and-stop	phantom strike, FR3 fingertips pressed on the bar at the cue (command path)	pass	-

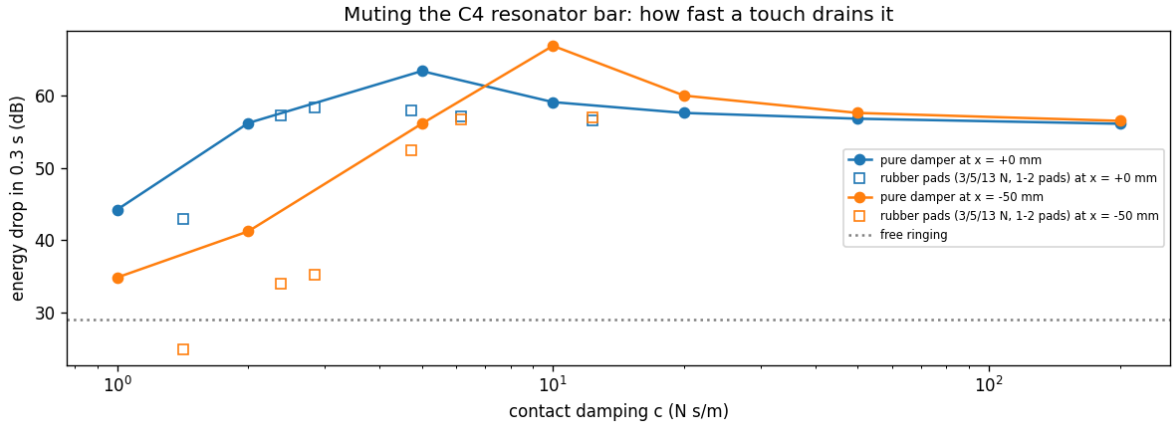
Strike-and-stop is a planned stop of stated length, judged on the struck bar's kinetic energy against the exact free continuation of its modal state at the cue (no extrapolated decay, no authored margin): 0.3271 s after the cue - one bar-box energy-exchange period - and for 0.1 s the bar must ring >= 10 dB below its free ring (half as loud). With no touch the two are equal (0.000 dB at every cue phase). The stop is planned because a touch within 0-100 ms of a cue of unknown delay gives phase-dependent verdicts.



Three of the recorded episodes the evaluator tests replay: the bar's kinetic energy (blue) against its free ring from the stroke (grey) and from the cue (orange), the judgement window shaded. FR3: the pads land 13.5 ms after the cue, 40 mm from the centre, 23.23 dB below the free ring.



Latency budget (no robot): the evaluator's measure for touches of a hand fixed in space landing -100..+200 ms around the cue, over eight cue phases. A firm or two-point touch near the centre within +-50 ms passes at every phase; a light single pad off-centre detunes the bar from its box and lengthens the ring.



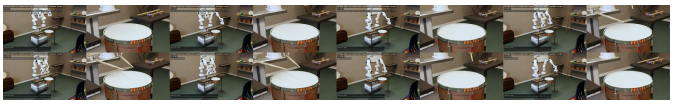
Mute bench (no robot): how fast a touch drains the C4 resonator bar, averaged over 0.4 s, for rubber pads at 3-13 N and pure dampers at the centre and 50 mm off it.

11 / The FR3 through the normal command path

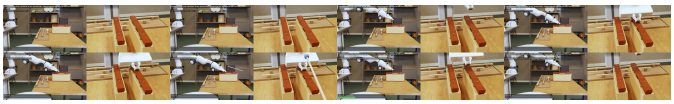
Scripted IK targets from privileged poses, sent as joint targets through the same rate limiter, servos and reflexes a policy gets. Most attempts fail. The strike-and-stop mute and take-sticks pass.

Experiment	Outcome	Checks
take sticks	pass	held_ready: y, no_strike: y, sticks_returned: y, nothing_dropped: y, robot_still: y
single tone	fail	one_stroke: n, clean: y, tone: y, level: n, nothing_dropped: y, robot_still: n
finger tap	fail	clean: n, no_rim: y, count: n, fingertip: y, tempo: n, level: n, nothing_dropped: y, robot_still: y
strike and stop	pass	struck_once: y, no_strike_after_cue: y, stopped: y, rang_until_cue: y, nothing_dropped: y, robot_still: y
pentatonic setup	fail	removed_to_tray: n, others_seated: y, nothing_dropped: y, robot_still: y
steady beat	fail	clean: n, no_rim: n, count: n, nothing_dropped: y, robot_still: y

The FR3's position interface decelerates at most 10 rad/s^2 per joint, so after a stroke the hand needs about 4 cm to stop and the rebounding stick is pressed back onto the head (robot_strokes: re-contacts and 117 ms presses). In the steady-beat attempt 0 of its 4 strike clusters are single and clean; the stick turns in the pinch and its shaft meets the hoop (19 rim events). This frictional-fulcrum rollout is sensitive to small changes in the setup. That is a property of the robot's command interface, kept as task difficulty rather than tuned away; the G1 configuration plays the full dynamic range (next chapter).



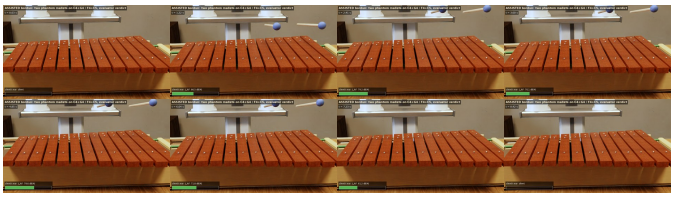
Take the sticks (evaluator pass).



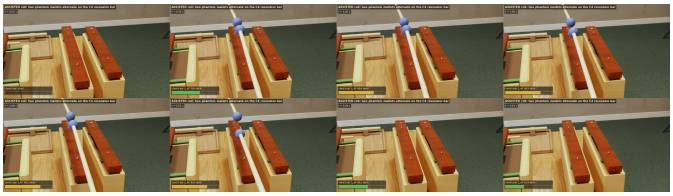
Strike-and-stop: phantom strike, the FR3's fingertips at the cue (evaluator pass).



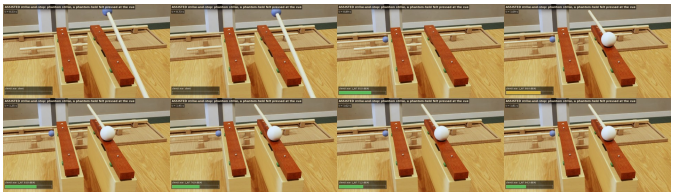
Single tone: a mallet stroke on the resonator bar that rattles (the hand cannot stop in time).



ASSISTED bordun: two phantom mallets on both bars together (evaluator pass).



ASSISTED roll on the C4 resonator bar (evaluator pass).



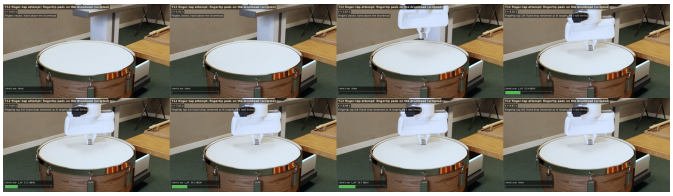
ASSISTED strike-and-stop: a phantom-held felt pressed at the cue.



ASSISTED dynamics: p, mf, f.



ASSISTED echo of the loudspeaker's rhythm.



FR3 fingertip taps: presses, not taps.

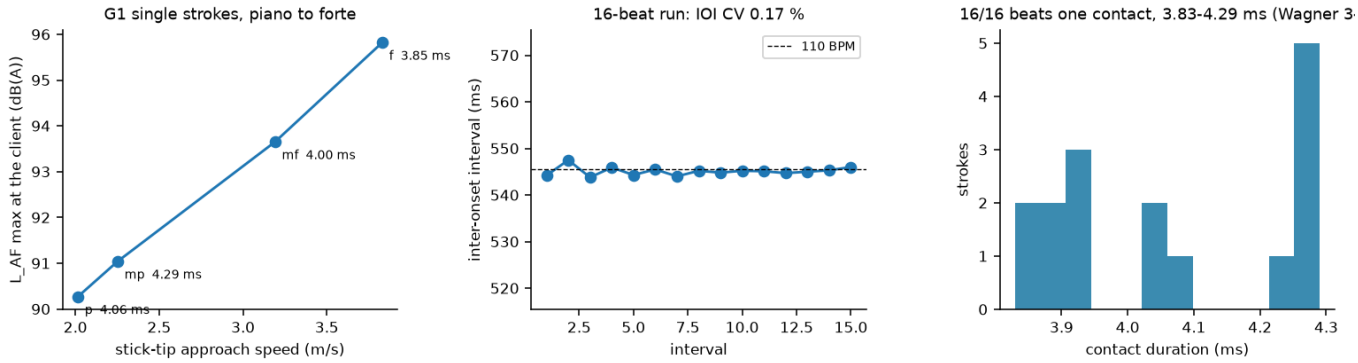


Steady-beat attempt: presses of 3-4 contacts; the stick turns in the pinch and its shaft meets the hoop.

12 / The G1 plays the drum

Joint targets through the normal command path. ASSISTED in g1_stick_strokes/g1_slowmo only: the stick is placed in the open hand and held by a phantom spring while the fingers close; the spring is then removed.

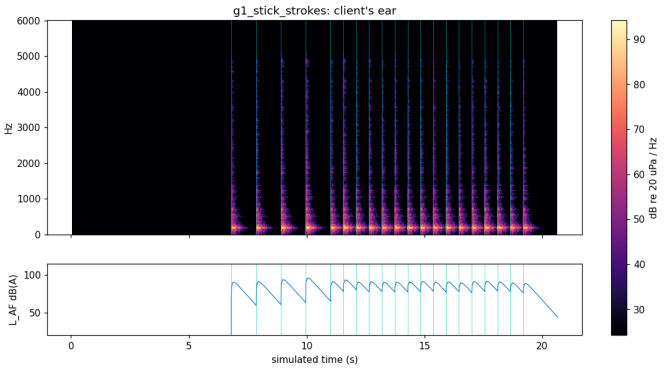
The stick is held by contact alone after the fixture is removed (held at the end: right). Elbow strokes with an up-stroke afterwards give one clean contact per stroke from 2.0 to 3.8 m/s; the 16-beat run at 110 BPM has inter-onset intervals within 0.17 % (CV). Rim contacts of the shaft with the hoop are detected and reported (ladder 0, beats 0).



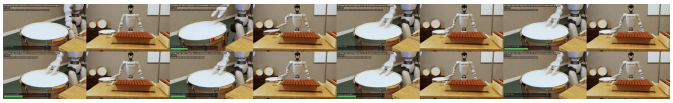
G1 stick strokes: level against speed (p..f), inter-onset intervals of the beat run, contact durations.



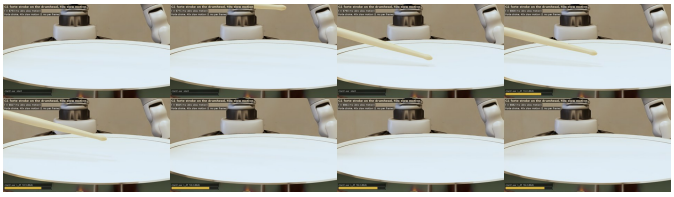
G1 stick strokes, contact sheet of the video.



Its spectrogram at the client.



G1 fingertip taps (Dex3 index and middle tips), evaluator checks: clean y, no_rim y, count y, fingertip y, tempo y, level y, nothing_dropped y, robot_still y



A forte stroke at 1000 frames/s (40x slow motion): approach, the 3.83 ms contact, rebound.

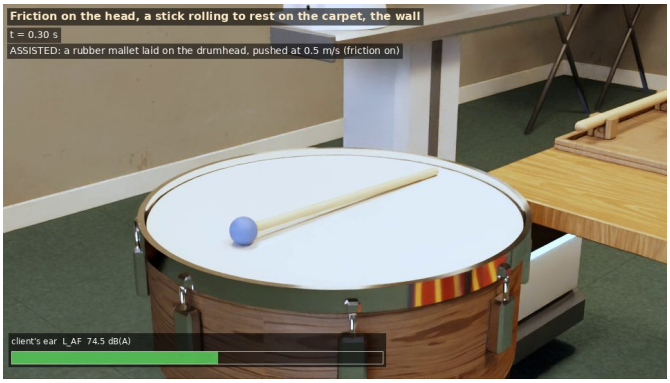
13 / Friction, hand surfaces, rim clicks and the room's boundaries

Each with its own experiment: friction_floor (assisted placements and pushes), hand_contacts (G1 through the command path), phantom_strokes (the rim click).

t (ms)	with friction: slid (mm)	speed (m/s)	frictionless: slid (mm)	speed (m/s)
10	3.85	0.3492	5.04	0.5055
40	12.36	0.1571	20.15	0.5047
70	15.25	0.0275	35.23	0.5023
100	15.38	0.0061	50.35	0.5053
130	15.39	0.0005	65.45	0.5025
160	15.4	0.0011	70.88	0.064
190	15.4	0.0005	68.96	0.0646
220	15.4	0.002	67.04	0.0659
250	15.4	0.0006	65.12	0.0643
280	15.4	0.0004	63.21	0.0623

A rubber mallet laid on the drumhead and pushed at 0.5 m/s stops after 15.4 mm under the vibro-contact's friction (peak tangential force 1.134 N); with friction disabled the same push glides 71 mm into the hoop and creeps back (an ablation). A stick rolled across the carpet at 0.66 m/s stops after 0.126 m; a stick thrown at 3 m/s toward the back wall bounces off it (closest approach y = -1.5934 m, wall at -1.6 m) and settles on the floor. The G1's hand pressed on the head brings finger-link proxies into contact (right_right_hand_index_1_link_proxy0, right_right_hand_index_1_link_proxy1); its slap is a logged contact heard at 73.7 dB(A). A stick on the hoop clicks: 2 hoop click(s) in phantom_strokes.

Silence under a resting load. Only the oscillating part of each mode radiates. The mallet lying still on the head: mean pressure -1.5e-07 Pa at the client, -1.1e-08 Pa at the station microphone, the policy's L_{Aeq}(100 ms) 0.0 dB(A). The G1's pressed hand: 1.8e-05 Pa static, AC 28.2 dB SPL half a second after the press.



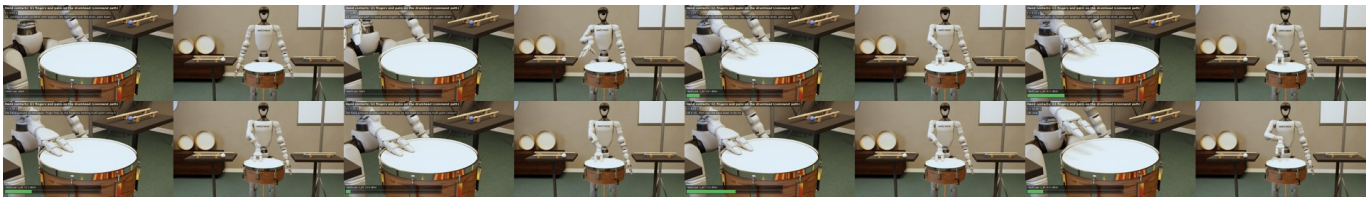
Friction test at 0.25 s: the pushed rubber mallet at rest on the head (friction).



Friction test, 0.6 s into the roll (it comes to rest after 0.13 m), in the floor camera's view.



Friction test, 1.2 s after the throw: the thrown stick has bounced off the back wall and settled (behind the rolled one).

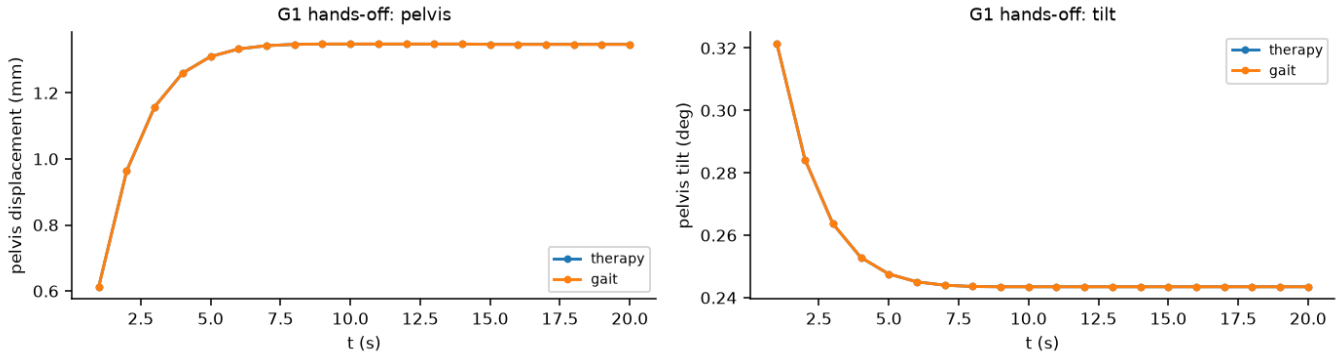


Hand contacts (drum close-up | the client's view): the G1 lays its fingers on the head, presses, lifts and slaps.

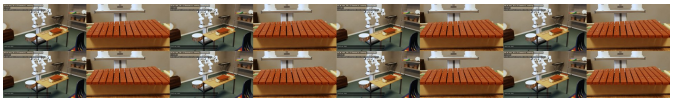
14 / Stationary behaviour and stability under legal actions

rest, g1_rest, layout sweeps, stress. Small settling is allowed; drift, NaN, infinities and explosions are failures.

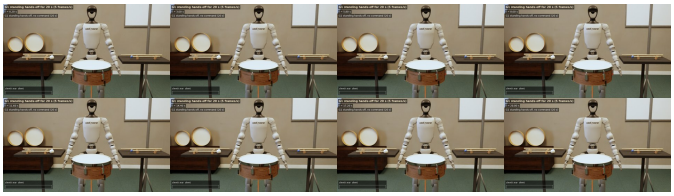
Run	Result
FR3 therapy, 20 s hands-off	max drift 0.0 mm, rotation 0.0396 deg, client mic peak 0.0 Pa, windows 0
FR3 gait, 20 s hands-off	max drift 0.0 mm, rotation 0.0 deg, client mic peak 0.0 Pa, windows 0
FR3 assessment, 20 s hands-off	max drift 0.0 mm, rotation 0.0396 deg, client mic peak 0.0 Pa, windows 0
FR3 pentatonic, 20 s hands-off	max drift 0.0 mm, rotation 0.0396 deg, client mic peak 0.0 Pa, windows 0
G1 therapy, 20 s hands-off	pelvis 1.347 mm, tilt 0.2435 deg, objects 0.0 mm, contacts 0
G1 gait, 20 s hands-off	pelvis 1.347 mm, tilt 0.2435 deg, objects 0.0 mm, contacts 0
Layout sweep FR3	36/36 layouts settle within 5 mm
Layout sweep G1	36/36 layouts settle, G1 up
Stress (FR3 + phantoms)	10 m/s stroke, stick resting on the head, 13-bar glissando, press, throw, 20 s of random legal commands: fault None, noslip fallbacks 0; fastest object over the run 6.583 m/s; every object inside the room at the end: True
G1 stress (legal targets)	slams on the head: 55 contacts, <= 43.2 N, robot up; 50 random targets for all 43 joints: falls (tilt 147.9 deg), fault None, finite True; knocked over by the falling robot: drum (peak 4.45 m/s, moved 1.28 m), stick_right (peak 5.506 m/s, moved 0.835 m); every object inside the room: True



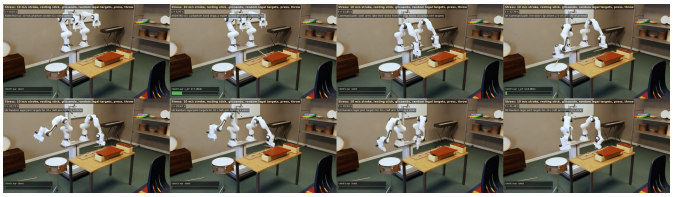
G1 standing hands-off for 20 s in two presets.



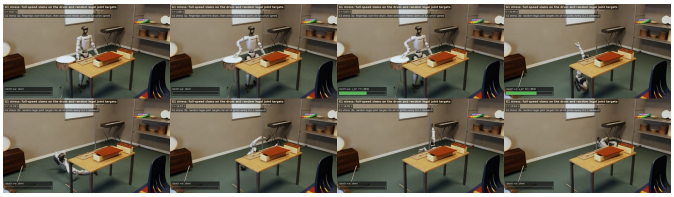
FR3 therapy room at rest, 20 s.



G1 at rest in the gait preset, 20 s.



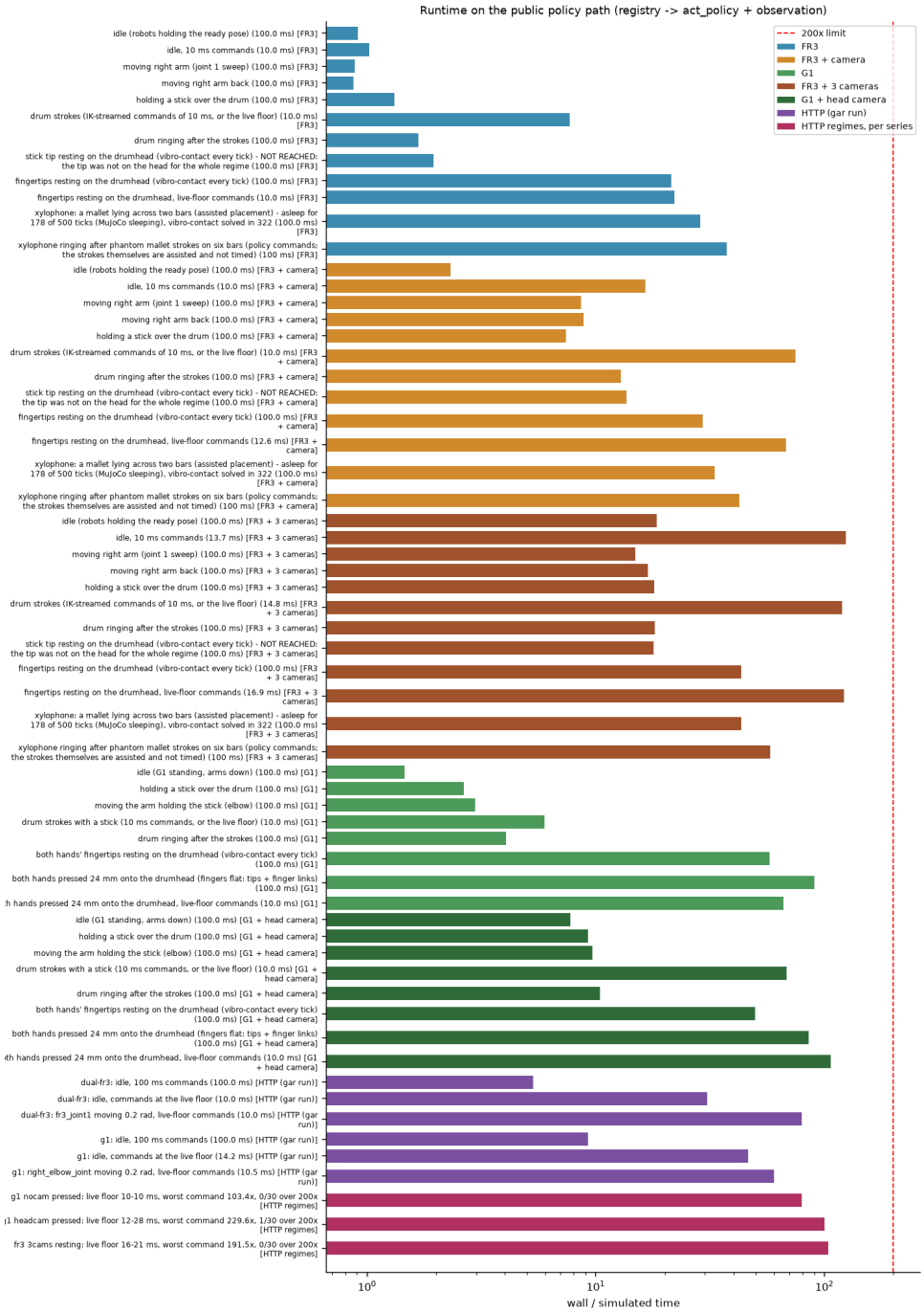
FR3 stress run.



G1 stress run: slams, then random legal targets until it falls.

15 / Runtime and interfaces

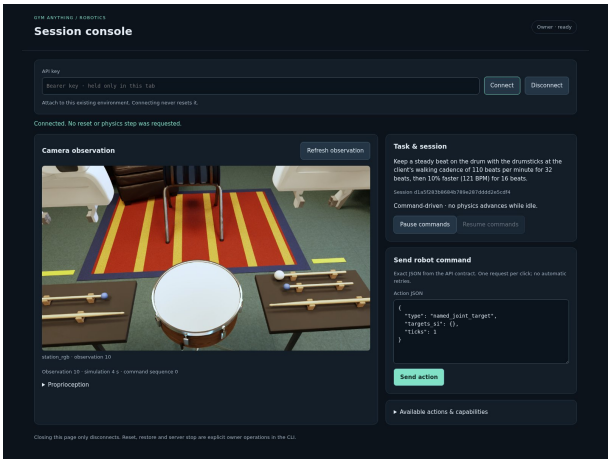
Runtime measured through the registry. The gar CLI, HTTP server and dashboard are checked for both robots under both keys.



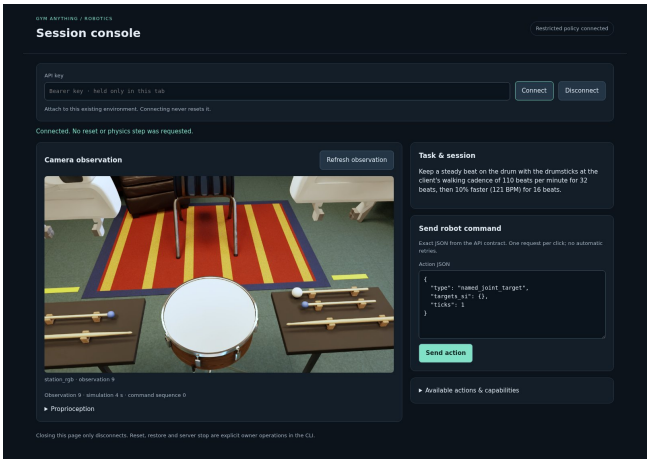
Wall/sim per regime and command length (log scale), measured on the public policy path (and over HTTP through gar run) while other jobs shared the machine; the policy path refuses commands shorter than the floor that follows the session's measured physics and per-command overhead.

Check	dual-fr3	g1
camera_png	pass	pass
dashboard_calls_200	pass	pass
dashboard_page_and_script_200	pass	pass
invalid_actions_refused	pass	pass
owner_save_restore_200	pass	pass
policy_actions_200	pass	pass
policy_audio_8000_samples	pass	pass

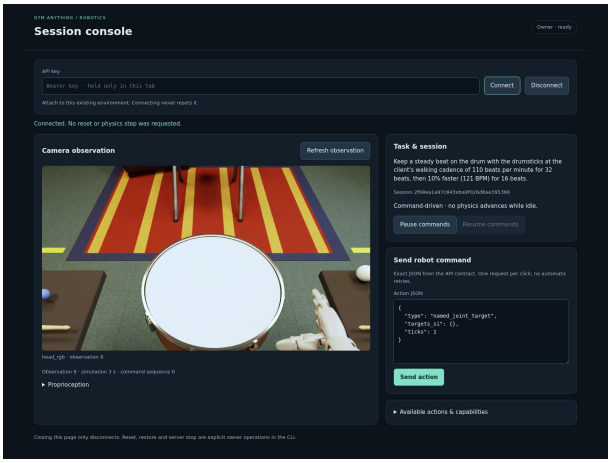
Check	dual-fr3	g1
policy_checkpoint_forbidden	pass	pass
policy_reads_200	pass	pass
policy_state_forbidden	pass	pass
restore_returns_tick	pass	pass
stopped	pass	pass



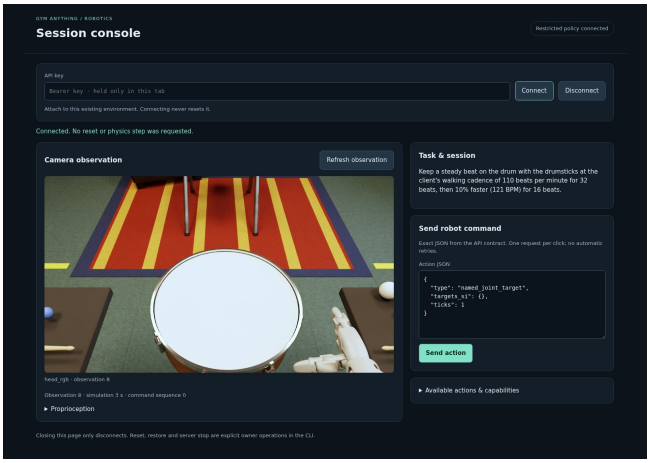
Dashboard, FR3 session (owner key: state, checkpoints, evaluation).



Dashboard, FR3 session (policy key: robot observations and actions only).



Dashboard, G1 session (owner key).

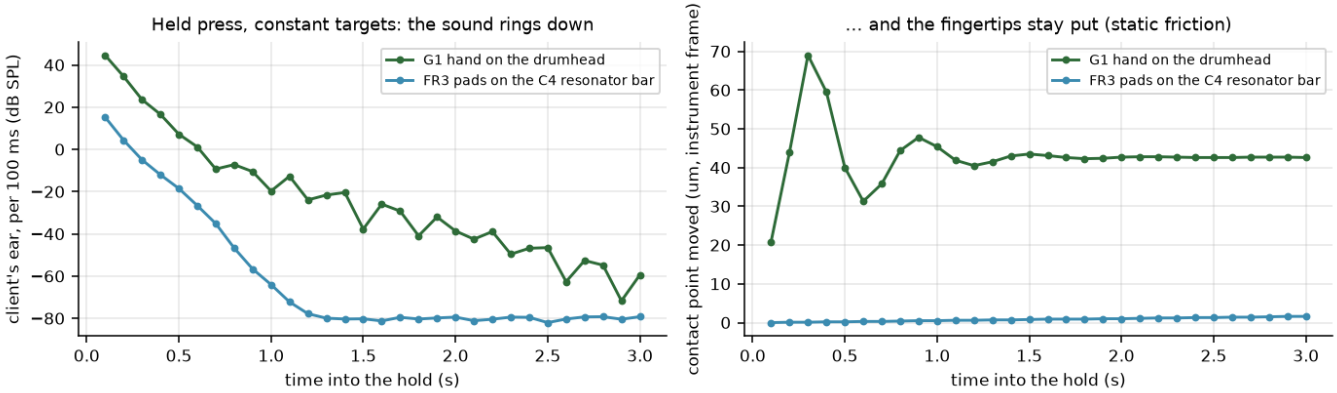


Dashboard, G1 session (policy key; head camera).

16 / Static friction, open contacts, bar losses, the resonator opening, per-command runtime

pressed_hold, stop_episodes, bench E1, resonator_sensitivity, command_path_seeds, runtime_http_regimes.

Held 3 s, constant targets	contact points moved (mm): whole hold / last second	client's ear, last second (dB SPL)	policy meters at the end (dB(A))
G1 hand pressed on the drumhead	0.043 / 0.0001	-38.9	L_AF 0.0, L_Aeq 0.0
FR3 pads pressed on the C4 resonator bar	0.002 / 0.0006	-79.2	L_AF 0.0, L_Aeq 0.0

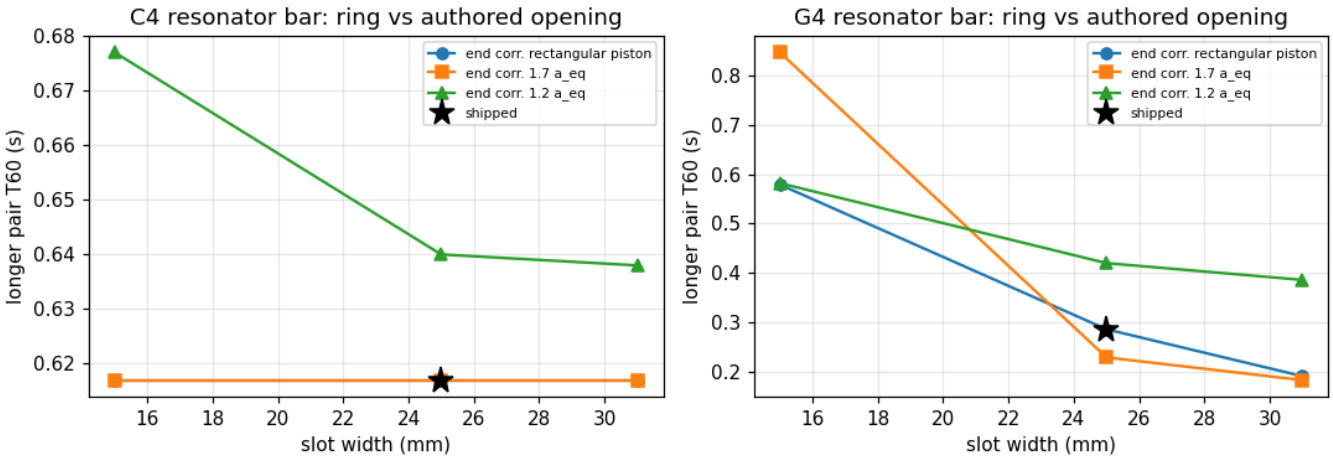


pressed_hold: every 100 ms of the 3 s hold, the client's-ear AC level and the largest displacement of a pressing contact point in the instrument's frame (G1 index/middle tips on the head; FR3 pads on the bar).

Static friction. A sticking contact is held by a tangential spring to an anchor on the surface (Mindlin's no-slip stiffness, 0.7 of the normal one, critically damped) and slips at mu G. A viscous sticking law carries a load only while moving, so a pressed G1 hand would creep and be heard. **Hand kinematics.** Inside an impact window the finger kinematics advance the body's orientation as well as its origin, so a moving pressed hand makes no 1 kHz tone. Lifting the pressed G1 hand: 0.4s: 46.5 dB(A) (200 Hz 2 %, 1 kHz 21 % of the sound); 2s: 51.5 dB(A) (200 Hz 1 %, 1 kHz 7 % of the sound); 0.4s targets every 5ms: 75.1 dB(A) (200 Hz 54 %, 1 kHz 0 % of the sound). Targets streamed every 5 ms step the servos at 200 Hz, next to the head's (0,1) mode.

Contacts still open at the evaluation count as segments ending now, with the largest vibration energy seen while they lasted: the felt that strikes the ringing bar 27 ms after the cue at 1.88 m/s and stays on it (still open: True) fails no_strike_after_cue (False). **Roll evenness** is judged on the strokes' approach speeds; the energy a stroke adds to a ringing bar depends on its phase.

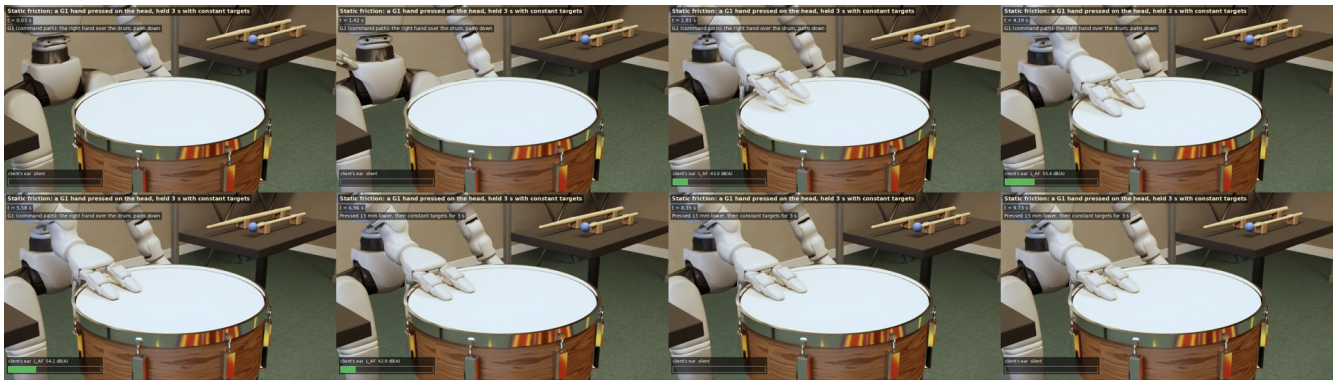
Bar	f1 (Hz)	T60 bare (s)	measured loss factor 0.0068 / 0.0045 / 0.0036 (s)	viscous law only (s)	on the xylophone (s)
C4	264	1.851	1.225 / 1.851 / 2.314	5.343	1.729
D4	296	1.651	1.093 / 1.651 / 2.064	4.25	1.3
F4	349	1.401	0.927 / 1.401 / 1.751	3.058	1.268
G4	392	1.247	0.825 / 1.247 / 1.559	2.424	1.235
A4	440	1.111	0.735 / 1.111 / 1.389	1.924	1.043
C5	520	0.94	0.622 / 0.94 / 1.175	1.377	0.78
E5	659	0.742	0.491 / 0.742 / 0.927	0.858	0.516
A5	880	0.481	0.368 / 0.555 / 0.694	0.481	0.366



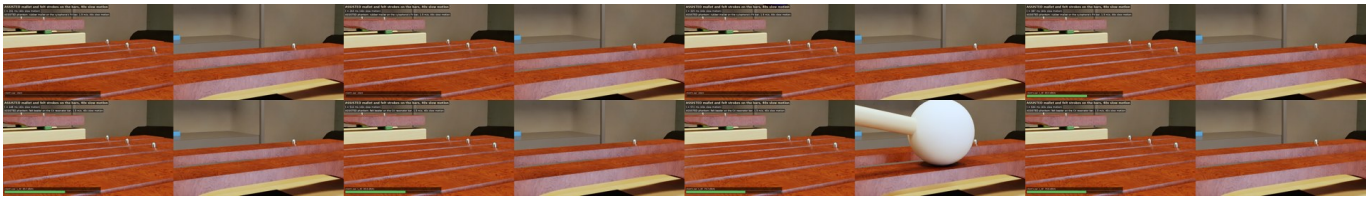
Resonator bars: the coupled bar-box pair's longer T60 against the authored slot width and end correction (each slot solved to tune the box to the bar's f1); the star is the shipped opening (25 mm, rectangular-piston end correction). Shipped: C4 slot 44.9 mm (14 % of the box), T60 0.604/0.617 s, exchange period 0.327 s; G4 slot 121.6 mm (43 % of the box), T60 0.124/0.286 s, exchange period 1.527 s.

Script	Seed	Outcome	Failed checks / error
FR3 strike-and-stop (phantom strike, fingertip mute)	1	fail	stopped
FR3 strike-and-stop (phantom strike, fingertip mute)	2	pass	-
FR3 strike-and-stop (phantom strike, fingertip mute)	3	fail	struck_once, stopped, robot_still
G1 finger-tap	1	fail	clean
G1 finger-tap	2	fail	clean, count, tempo, level

Regime over HTTP (policy key, live floor)	floors (ms)	wall/sim (series)	incl. caps + image GETs	worst command	commands > 200x
g1 nocam pressed	10-10	79.4	79.6	103.4	0/30
g1 headcam pressed	12-28	99.5	99.7	229.6	1/30
fr3 3cams resting	16-21	103.3	103.7	191.5	0/30



Pressed hold: the G1's hand pressed on the head and held with constant targets.



Mallet slow motion (ASSISTED, 1000 frames/s, 40x slow motion): a rubber mallet on the xylophone's F4 bar (left) and a felt beater on the C4 resonator bar (right), low cameras at the struck points.

17 / Limitations

What the environment does not model or has not shown.

- Absolute loudness: tasks use reference strokes replayed through the same model. One absolute check: a 10 m/s accent peaks at 130.29 dB (f, 6 m/s: 125.84 dB) at a seated player's head against 130.8 dB L_peak measured there for a snare with the snares off (Jaroszewski, Rogowski & Rakowski 2000, Table 1).

- Estimated parameters without a direct source: drumhead coating mass, contact laws of felt, rubber and fingertip (the Franka pad as rubber E 3 MPa, R 20 mm), resonator coupling, friction coefficients other than bare PET film (the coated head's included), the rim-click radiation fraction, room construction, xylophone box widths, the bar reference speeds; the resonator-bar boxes follow the product's outer size and mass, but slot width, box lengths between c1 and h1 and their decays (C4 T60 0.62 s, G4 0.29 s) are not checked against a recording.

- The resonator bars' decay rests on the authored opening (resonator_sensitivity: G4 0.18-0.85 s, C4 0.62-0.68 s but its bar-box exchange period, strike-and-stop's judgement delay, 0.30-5.3 s over plausible slots); the G4 box is past the lumped Helmholtz model (0.33 wavelengths long, slot over 43 % of it).

- The bars' damping floor is a measured loss factor of rosewood marimba bars (Hill 2011), not of an Orff alto xylophone; the low bars' decay on the instrument (C4 1.7 s) is not checked against a recording.

- Command-path successes are per seed: the FR3 strike-and-stop and G1 finger-tap scripts pass on their seed 0; other seeds are reported (command_path_seeds).

- Bars meet each other and the frame through 7 mm collider slabs (the recessed top lets fingers reach the vibro-contact first): tilted bars can overlap their neighbours' visible bodies by a few millimetres.

- The command floor is a forecast from the last 32 commands (x1.25): a command whose own render, disk or physics time jumps past them can exceed 200x; series averages stay below it.

- The shell, stand and table do not vibrate as modal structures; a rigid hit on them radiates an authored transient scaled by impact energy (an estimate). The open shell's back face does not radiate above the head.

- Strike-and-stop is a planned stop of stated length, judged on the bar's vibration energy against its exact free ring (privileged); a reactive stop at human reaction time on an unannounced cue does not pass, by design.

- The FR3 cannot stop its hand within a stroke's rebound under its position interface: loud clean strokes are out of its reach; the tasks keep that difficulty. It passes take-sticks and the stop of strike-and-stop (the latter on assessment seeds 0 and 2 of 0-3).

- The G1's stick grasp in the evidence starts from an assisted placement; a Dex3 pick of the stick from the trap table is not demonstrated. The G1 has no balance controller; it stands on its servos.

- Muting is checked against the model's own bench and latency probe, not a measurement of a muted resonator bar. Resting rubber-mallet losses are not modelled (Rucz's elastic law).

- The stick's contacts (3.7-4.7 ms) sit inside Wagner's 3-6 ms but below Dahl's 4.5-8 ms in total, without Wagner's rim-reflection re-contacts.

- Sound is a modal point-source model with first-order images and a statistical tail, not a wave solve; the furniture colliders are bounding boxes.

- Resting and pressed robot contacts are the expensive physics (every pair solved per audio sample). The command floor follows the session's measured cost, so the 200x budget holds on average while physics alone stays below it.

- Not built: the preschool frame-drum task; a client figure in the chair; a single-panda configuration.

18 / Sources

Each source with what it supports.

- [S14] Rucz, Ulveczki, Angster, Miklos, Simulation of mallet percussion instruments by a coupled modal vibroacoustic finite element model, JASA 149(5):3200-3212 (2021) - partials, masses, damping and resonators of an Orff alto xylophone's bars; mallet contact.

- [S12] Wagner 2006 (14 x 5 in drum, Remo Ambassador coated head: modes at three tensions, stick contact and re-contacts); [S10] Dahl 2004, [S11] Dahl et al. 2011 - stroke heights, speeds, forces and levels.

- Chaigne and Doutaut 1997; Bork; Soares, Antunes and Debut - bar/mallet impact models and resonator tubes.

- Hunt and Crossley 1975 - dissipative impact law; collision numerics literature for implicit contact.

- Fletcher and Rossing, The Physics of Musical Instruments - membranes, bars, radiation.

- Franka Robotics FR3 datasheet and libfranka rate-limiting source - joint velocity, acceleration and jerk limits, collision behaviour.

- Unitree G1 specification (head microphone array, D435 camera); the pinned G1 URDF; whiteboard and barcode_scanning G1 qualification (servo profile, 0.5 ms step, gravity feed-forward).

- Remo, Vic Firth and Studio 49 product data - drum, stick and instrument dimensions and materials.

- miniDSP UMIK-1 (measurement microphone); IEC 61672 (A-weighting, fast time weighting); ANSI S12.60 (classroom acoustics); EN 12464-1 (illuminance).

- Orff-Schulwerk and music-therapy practice sources (instrument choice, pentatonic set-up, rhythmic auditory stimulation).

- Poly Haven CC0 models and textures (pinned by SHA-256 in visual_assets.json).

- [S45] rubber damping reference tables (natural rubber tan delta 0.05-0.15, silicone 0.10-0.30) - pad loss factor; [S46b] DuPont Teijin Mylar datasheets (ASTM D1894 COF 0.40 dynamic, 0.50 static) - stick on head friction.

- [S47] Hoffman et al. 2003, wheelchair coasting (carpet adds C_RR 0.0118); [S48] Sauret et al. 2012 (loop pile carpet 5.4 mm) - the floor's rolling resistance.

- [S49] Owens Corning Noise Control Design Guide (absorption of painted gypsum board on studs) - wall absorption.

- [S50] Jaroszewski, Rogowski, Rakowski, Sound pressure levels in emission of percussion instruments during training sessions, Archives of Acoustics 25(3):283-289 (2000), Table 1 (snare, snares off, 130.8 dB L_peak at the player's head) - the absolute-level check; [S51] Valderrama, Arenas, Cardenas, IJAV 30(4) (2025) (kit snare 134.8 dB at the ear canal, moderate force).

- [S52] Rempel et al. 1994, J Biomech 27(8) (fingertip speeds of keystrokes 0.3-0.7 m/s); [S53] Hajian and Howe 1997, J Biomech Eng 119(1) (fingertip equivalent mass 21.5 g) - the finger-tap reference tap.

- [S54] Repp 2005, Psychon Bull Rev 12(6) (anticipatory tapping to a predictable beat); [S55] ISO 532-1:2017 (loudness doubles per 10 phon) - strike-and-stop's timing and 10 dB.

- [S27b] Studio 49 KB/AX resonator bar dealer datasheets (32-27 cm x 5 x 11 cm, 0.40 kg) - the boxes.

- [S56] Hill 2011, Exponential time decay constants of marimba bars, University of Illinois REU report (rosewood bars in situ, loss factor 0.0036-0.0068, mean 0.0045) - the bars' damping floor; [S57] Mindlin 1949, J Appl Mech 16, and Johnson, Contact Mechanics 7.2 (no-slip tangential stiffness 2(1-nu)/(2-nu) of the normal) - static friction.